Thinking with algebra

L1-AL1 Level 1 Foundations · Algebra

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Using a letter for an unknown number, turning a word problem into an equation, solving the equation, and checking the answer against the question.
By the end you can choose a letter for an unknown and write every other quantity in terms of that letter; turn money, sharing and age problems into an equation and solve the equation; undo a process step by step when only its result is known.
Prerequisites: none. **Contests:** JMC, Junior Kangaroo, JMO, AMC 8. **Time:** about 60 minutes.

1 Lesson

Priya has 31 stamps and Omar has 13. How many stamps must Priya give Omar so that they have the same number?

Many people answer 18, because Priya has $31 - 13 = 18$ more than Omar. Test this answer. Priya would then have 13 stamps and Omar would have 31. They have swapped, and the gap is still 18. So 18 is wrong.

Instead, say that Priya gives Omar x stamps. Then she has $31 - x$ stamps and he has $13 + x$, and we want these to be equal:

$$31 - x = 13 + x.$$

Adding x to both sides gives $31 = 13 + 2x$, so $2x = 18$ and $x = 9$. As a check, Priya has $31 - 9 = 22$ and Omar has $13 + 9 = 22$. Priya must give Omar 9 stamps.

The equation also shows what went wrong. Each stamp that moves makes Priya's pile one smaller and Omar's pile one larger, so each move closes the gap by 2, and a gap of 18 is closed by 9 stamps.

This unit covers that method, and two kinds of problem that need extra care, namely problems about ages and processes that have to be worked backwards.

1.1 Letters stand for numbers

A letter in algebra stands for a number that we do not know yet. Usually the best choice is to give a letter to the quantity the question asks for, or to the smallest quantity in the problem, and then write every other quantity in terms of that letter. The phrases below come up again and again.

- "7 more than p" is $p + 7$, and "7 less than p" is $p - 7$.
- "Three times as many as b" is $3b$, and "a third of x" is $\frac{x}{3}$.
- "Is", "equals" and "will be the same as" become the sign $=$.

Decide on the units at the start (pounds or pence, years or months) and use the same units throughout.

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1.2 Forming and solving an equation

The method.

1. Say in words what your letter stands for.
2. Write every other quantity in terms of that letter.
3. Find a fact in the question that you have not used yet. Write this fact as an equation, that is, two expressions for the same quantity, joined by $=$.
4. Solve the equation by doing the same thing to both sides.
5. Check the answer against the words of the question.

Step 4 works because the two sides of an equation are the same number. If we add, subtract, multiply or divide both sides by the same thing (but never divide by zero), they are still equal. For example, from $5c + 3 = 6c - 4$ we subtract $5c$ from both sides to get $3 = c - 4$, then add 4 to both sides to get $c = 7$.

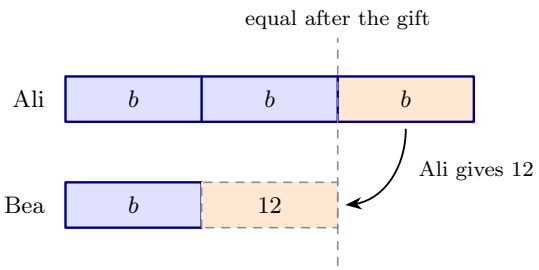
In step 5, check against the words, not against your equation. If you made a mistake when writing the equation, the wrong answer will still fit the equation, and only the words will show the mistake.

Example 1.
Ali has three times as much money as Bea. If Ali gives Bea £12, they will have the same amount. How much money do they have altogether?

Solution.
Bea has the smaller amount, so say Bea has b pounds. Then Ali has $3b$ pounds. After the gift, Ali has $3b - 12$ pounds and Bea has $b + 12$ pounds, and these are equal:

$$3b - 12 = b + 12.$$

Subtracting b from both sides gives $2b - 12 = 12$, and adding 12 to both sides gives $2b = 24$, so $b = 12$. Bea has £12 and Ali has £36, so together they have £48.
Checking against the words, £36 is three times £12, and after the gift each has £24.
As with the stamps, the gift reduces the gap between them by £24, not £12, because Ali loses £12 and Bea gains £12. The bar model below shows the same thing. After the gift the two amounts meet at $2b$, so the part Ali gives away is one whole b, and hence $b = 12$.



Example 2.
A teacher shares a bag of sweets among the children in her class. If each child gets 5 sweets,

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there are 3 sweets left over. If each child gets 6 sweets, she is 4 sweets short. How many children are there, and how many sweets?

Solution.
There are two unknowns, the number of children and the number of sweets. But the number of sweets is described twice, once for each way of sharing, and that is the fact to use in step 3.
Say there are c children. Giving 5 each uses $5c$ sweets and leaves 3, so there are $5c+3$ sweets. Giving 6 each would need $6c$ sweets, and she has 4 fewer than that, so there are $6c-4$ sweets. The two counts are for the same bag, so

$$5c+3=6c-4.$$

Subtracting $5c$ from both sides gives $3=c-4$, so $c=7$. There are 7 children and $5\times 7+3=38$ sweets.
As a check, 7 children with 5 each use 35 sweets, leaving 3. With 6 each they would need 42, which is 4 more than 38.
The same argument works in words. Giving every child one extra sweet uses up the 3 that were left over and needs 4 more, so one sweet per child is $3+4=7$ sweets in total, and there are 7 children. Subtracting $5c$ from both sides of the equation is exactly this step.

1.3 Ages

Age problems use one fact over and over. Everyone gets one year older each year. So in n years' time, add n to *every* age, and n years ago, subtract n from every age. Hence the difference between two people's ages never changes.

Example 3.
Mia is 11 and her mother is 39. (a) In how many years will her mother be twice as old as Mia? (b) How many years ago was her mother five times as old as Mia?

Solution.
(a) In n years Mia will be $11+n$ and her mother will be $39+n$. We need $39+n=2(11+n)=22+2n$. Subtracting n and then 22 from both sides gives $n=17$. As a check, Mia will be 28 and her mother 56, which is twice 28.
(b) Here we go back m years, so Mia was $11-m$ and her mother was $39-m$. We need $39-m=5(11-m)=55-5m$. Adding $5m$ to both sides gives $39+4m=55$, so $4m=16$ and $m=4$. Also, four years ago Mia was 7 and her mother was 35, which is five times 7.
The difference gives a second method. The difference is always $39-11=28$. When the mother is twice Mia's age, the difference equals Mia's age, so Mia is 28, which is 17 years from now. When the mother is five times Mia's age, the difference is four times Mia's age, so Mia is $28\div 4=7$, which was 4 years ago.

1.4 Working backwards

Some questions describe a process step by step and tell you only how the process ends. Then start from the end and undo the steps in reverse order. Each step is undone by its opposite, so adding is undone by subtracting and multiplying by dividing. Be careful with fractions. If someone gives away half of what they have, they keep half, so you undo that step by doubling. If they give away a third, they keep two thirds, so what is left is two of three equal parts.

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Example 4.
Sam had some stickers. He gave half of them, plus one more, to his brother. He then gave a third of the stickers he had left to his sister, and he had 10 stickers left. How many stickers did Sam start with?

Solution.
We know how the process ends, so we work backwards from the end.
Giving his sister a third left Sam with two thirds of what he had, and two thirds of that amount is 10. So one third is 5, and before he met his sister Sam had 15 stickers.
Before that, he gave his brother half of his stickers and one more. Had he given away only the half, he would have kept one more sticker, which is 16. So 16 is half of his starting number, and he started with 32. (With a letter, if he started with x then he kept $\frac{x}{2} - 1$, so $\frac{x}{2} - 1 = 15$, which gives $\frac{x}{2} = 16$ and $x = 32$.)
Check forwards. Half of 32 plus one is 17, leaving 15. A third of 15 is 5, leaving 10. Sam started with 32 stickers.
Note that the step "half plus one" is undone by adding one and then doubling, in that order. Doubling first and then adding one gives 31, which is wrong.

Common traps.

- Mixing units, such as pounds in one place and pence in another.
- Reading "4 short" as $+4$. Being short means having fewer, so the value is -4.
- In age problems, adding the years to only one person's age.
- Undoing steps in the wrong order, or undoing "gives away half" by halving.
- Solving for the letter and forgetting what the question asked for, such as a total.

1.5 How to spot this in a contest

1. If the question describes quantities by how they compare ("three times as many", "£7 more than"), give a letter to the smallest one and write the others in terms of that letter.
2. If the same quantity is described in two ways (two ways of sharing, or before and after a gift), write both expressions and set them equal.
3. In age problems, add or subtract the same number of years from every age, and use the fact that age differences never change.
4. If a process is described and only the end result is given, work backwards, undoing each step.
5. On a multiple-choice paper you can test the options in the words of the question. This is a fair way to check an answer, and sometimes the quickest method.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ When I double a number and then subtract 9, I get the same answer as when I add

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14 to the number. What is the number?

Problem 2 ⋆ Bilal, Chloe and Dev share £90. Bilal gets £6 more than Chloe, and Dev gets twice as much as Chloe. How much does Dev get?

Problem 3 ⋆⋆ Think of a number. Add 5, multiply the result by 4, subtract 8, divide by 4, and finally subtract the number you first thought of. Explain why the answer is always 3, whatever number you start with.

Problem 4 ⋆⋆ Some pupils go on a trip in minibuses. If 12 pupils sit in each minibus, then 4 pupils have no seat. If instead 14 pupils sit in each minibus, then one minibus has only 6 pupils and all the others are full. How many pupils are going on the trip?

Problem 5 ⋆⋆⋆ Four years ago, Jo's age was one third of her aunt's age. In eight years' time, Jo's age will be half of her aunt's age. How old is Jo now?

Problem 6 ⋆⋆⋆ Amy, Ben and Cal each have some marbles. First Amy gives Ben as many marbles as Ben already has. Then Ben gives Cal as many marbles as Cal already has. Finally Cal gives Amy as many marbles as Amy has at that moment. Now each of them has 16 marbles. How many marbles did Amy have at the start?

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AMC 8 2022 Problem 5** ⋆
   Work out each age today, one at a time
2. **UKMT Junior Kangaroo 2017 Question 5** ⋆
   The age gap between the two never changes
3. **UKMT JMC 2015 Question 12** ⋆
   Call the weight w and write the sentence as an equation
4. **UKMT Junior Kangaroo 2019 Question 4** ⋆⋆
   Let Evie's age now be a letter and write both ages in terms of that letter
5. **UKMT JMC 2018 Question 14** ⋆⋆
   Give a letter to the smaller number for each farmer
6. **UKMT Junior Kangaroo 2017 Question 8** ⋆⋆
   Call the final number in each tree n and work back to the start
7. **UKMT Junior Kangaroo 2018 Question 16** ⋆⋆
   Start from the fourth shelf and work backwards
8. **AMC 8 2022 Problem 11** ⋆⋆
   Count the bites from the number of pieces, then work backwards
9. **AMC 8 2017 Problem 17** ⋆⋆
   Write the number of coins in two ways using the number of chests

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10. **AMC 8 2010 Problem 21** ⋆⋆⋆
    Work backwards from the pages left, one day at a time

11. **UKMT JMO 2018 Question B1** ⋆⋆⋆
    Compare how fast Polly's age and the sum of the children's ages increase

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4 Hints

1. Call the number x and write each side of the sentence in terms of x.
2. Chloe's share is the one the others are compared with, so give her share a letter.
3. Trying a few numbers shows that the answer is 3 but does not explain why. Call the starting number n and follow this number through each step.
4. Let m be the number of minibuses and write the number of pupils in two different ways.
5. The difference between their ages never changes. What fraction of that difference was Jo's age four years ago, and what fraction will that age be in eight years?
6. Work backwards from the end. When someone receives as many marbles as they already have, their number doubles, so before the gift they had half as many.

5 Solutions

Problem 1

When I double a number and then subtract 9, I get the same answer as when I add 14 to the number. What is the number?

Let the number be x. Doubling and subtracting 9 gives $2x - 9$, and adding 14 gives $x + 14$. These are equal, so $2x - 9 = x + 14$. Subtracting x from both sides gives $x - 9 = 14$, and adding 9 gives $x = 23$. Check that $2 \times 23 - 9 = 37$ and $23 + 14 = 37$.

Answer: 23

Problem 2

Bilal, Chloe and Dev share £90. Bilal gets £6 more than Chloe, and Dev gets twice as much as Chloe. How much does Dev get?

Let Chloe get c pounds. Then Bilal gets $c + 6$ pounds and Dev gets $2c$ pounds. Together they get $c + (c + 6) + 2c = 4c + 6$ pounds, so $4c + 6 = 90$. Subtracting 6 gives $4c = 84$, so $c = 21$. Dev gets twice this, which is £42. Check that the shares are £27, £21 and £42, which add up to £90. Bilal's share is £6 more than Chloe's, and Dev's is twice Chloe's.

Answer: £42

Problem 3

Think of a number. Add 5, multiply the result by 4, subtract 8, divide by 4, and finally subtract the number you first thought of. Explain why the answer is always 3, whatever number you start with.

Call the starting number n. Adding 5 gives $n + 5$. Multiplying by 4 gives $4(n + 5) = 4n + 20$. Subtracting 8 gives $4n + 12$. Dividing by 4 gives $n + 3$, since both $4n$ and 12 are divided by 4. Finally, subtracting the starting number n leaves 3.

Every step was done to a general number n, so the argument works for every starting number at once, and the answer is always 3. Testing examples, however many, would not prove this. The letter is what allows us to deal with all numbers together.

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Problem 4

Some pupils go on a trip in minibuses. If 12 *pupils sit in each minibus, then* 4 *pupils have no seat. If instead* 14 *pupils sit in each minibus, then one minibus has only* 6 *pupils and all the others are full. How many pupils are going on the trip?*

Let there be m minibuses. In the first arrangement the minibuses hold $12m$ pupils and 4 more are left over, so there are $12m + 4$ pupils. In the second arrangement $m - 1$ minibuses are full with 14 pupils each and one has 6, so there are $14(m - 1) + 6 = 14m - 8$ pupils.

These are the same pupils, so $12m + 4 = 14m - 8$. Subtracting $12m$ from both sides gives $4 = 2m - 8$, so $2m = 12$ and $m = 6$. The number of pupils is $12 \times 6 + 4 = 76$.

As a check, 6 minibuses of 12 hold 72, leaving 4 without a seat. Five full minibuses of 14 hold 70, and the sixth holds the remaining 6.

Answer: 76

Problem 5

Four years ago, Jo's age was one third of her aunt's age. In eight years' time, Jo's age will be half of her aunt's age. How old is Jo now?

Let Jo be j and her aunt be a now. Four years ago they were $j - 4$ and $a - 4$, so $a - 4 = 3(j - 4) = 3j - 12$, and adding 4 to both sides gives $a = 3j - 8$. In eight years they will be $j + 8$ and $a + 8$, so $a + 8 = 2(j + 8) = 2j + 16$, and subtracting 8 gives $a = 2j + 8$. Both expressions equal a, so $3j - 8 = 2j + 8$, and subtracting $2j$ and adding 8 gives $j = 16$. Then $a = 40$.

As a check, four years ago Jo was 12 and her aunt was 36, which is three times 12. In eight years Jo will be 24 and her aunt 48, which is twice 24.

The difference gives a second method. Call the age difference d, which never changes. Four years ago the aunt's age was three times Jo's, so the difference was twice Jo's age, and Jo was $\frac{d}{2}$. In eight years the aunt's age will be twice Jo's, so the difference will equal Jo's age, and Jo will be d. These two ages of Jo are 12 years apart, so $d - \frac{d}{2} = 12$, which means $\frac{d}{2} = 12$. So Jo was 12 four years ago and is 16 now.

Answer: 16

Problem 6

Amy, Ben and Cal each have some marbles. First Amy gives Ben as many marbles as Ben already has. Then Ben gives Cal as many marbles as Cal already has. Finally Cal gives Amy as many marbles as Amy has at that moment. Now each of them has 16 *marbles. How many marbles did Amy have at the start?*

We work backwards from the end, where each has 16 marbles.

The last step was Cal giving Amy as many as she had, which doubled Amy's number. So before that step Amy had 8, and Cal had given her 8, so Cal had $16 + 8 = 24$. Ben was not involved and had 16. So before the last step the numbers were Amy 8, Ben 16, Cal 24.

The step before was Ben giving Cal as many as Cal had, which doubled Cal's number. So Cal had 12 before that step, and Ben had $16 + 12 = 28$. The numbers were Amy 8, Ben 28, Cal 12.

The first step was Amy giving Ben as many as Ben had, which doubled Ben's number. So Ben had 14 before that step, and Amy had $8 + 14 = 22$. At the start the numbers were Amy 22, Ben 14, Cal 12.

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Check forwards. Amy gives Ben 14, giving 8, 28, 12. Ben gives Cal 12, giving 8, 16, 24. Cal gives Amy 8, giving 16, 16, 16. The total stays at 48 throughout, as expected, since marbles only move between the three of them. Amy started with 22 marbles.

Answer: 22